

More Consequences of MVT

Def A function f is increasing on an interval I if $\forall a, b \in I$ w/ $a \leq b$, then $f(a) \leq f(b)$; decreasing if whenever $a \leq b$, $f(a) \geq f(b)$

Cor If $f'(x) > 0 \quad \forall x \in I$, then f is strictly increasing on I .

Pf Let $a, b \in I$, $a < b$ (wts: $f(a) < f(b)$)

By MVT, $\exists x \in [a, b]$ s.t. $f'(x) = \frac{f(b) - f(a)}{b - a} > 0$

Since $b - a > 0$ by our choice, we must have $f(b) > f(a)$.



Prop If $f(x) = x^n$, then $f'(x) = nx^{n-1}$

Pf. Let's use binomial thm, by definition, $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$

$$\text{Binomial thm: } (a+b)^n = \sum_{i=0}^n \binom{n}{i} a^i b^{n-i} = \frac{n!}{i!(n-i)!}$$

$$(x+h)^n - x^n = \sum_{i=0}^n \binom{n}{i} x^i h^{n-i} - x^n$$

$$= \binom{n}{0} h^n + \binom{n}{1} h^{n-1} x + \dots + \binom{n}{n} x^n - x^n$$

$$= h^n + nh^{n-1}x + \dots + nhx^{n-1}$$

$$\begin{aligned} \text{Then } \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} &= \lim_{h \rightarrow 0} (h^{n-1} + nh^{n-2}x + \dots + nx^{n-1}) \\ &= nx^{n-1} \end{aligned}$$

Rules of Differentiation

Sums If f, g differentiable at a , then $(f+g)$ is also differentiable at a & $(f+g)'(a) = f'(a) + g'(a)$

Pf. $(f+g)'(a) = \lim_{h \rightarrow 0} \frac{(f+g)(a+h) - (f+g)(a)}{h}$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) + g(a+h) - f(a) - g(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} + \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h}$$

$$= f'(a) + g'(a)$$

Products If f, g differentiable at a , then $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.

Pf By definition of the derivative,

$$(fg)'(a) = \lim_{h \rightarrow 0} \frac{(fg)(a+h) - (fg)(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h)g(a+h) - f(a)g(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h)g(a+h) - f(a+h)g(a) + f(a+h)g(a) - f(a)g(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h)[g(a+h) - g(a)] + g(a)[f(a+h) - f(a)]}{h}$$

exists b/c g diff $\rightarrow$ split

$$= \lim_{h \rightarrow 0} \left[f(a+h) \frac{g(a+h) - g(a)}{h} + g(a) \frac{f(a+h) - f(a)}{h} \right]$$

exists b/c f diff $\Rightarrow$ fcts.

$$= f'(a)g(a) + f(a)g'(a)$$

□

Quotient

① If g is differentiable at a , $g(a) \neq 0$, then $1/g$ is diff. at a .

$$\left(\frac{1}{g}\right)'(a) = \frac{-g'(a)}{[g(a)]^2}$$

② If f, g diff at a , $g(a) \neq 0$, then f/g diff at a .

$$\left(\frac{f}{g}\right)'(a) = \frac{g(a)f'(a) - f(a)g'(a)}{[g(a)]^2}$$

Pf ① By definition, $\left(\frac{1}{g}\right)'(a) = \lim_{h \rightarrow 0} \frac{\frac{1}{g(a+h)} - \frac{1}{g(a)}}{\frac{1}{h}}$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{g(a+h)} - \frac{1}{g(a)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{g(a) - g(a+h)}{hg(a)g(a+h)}$$

$$= -\frac{1}{g(a)} \lim_{h \rightarrow 0} \left(\frac{g(a+h) - g(a)}{h} \cdot \frac{1}{g(a+h)} \right)$$

$$= -\frac{1}{g(a)} \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} \cdot \lim_{h \rightarrow 0} \frac{1}{g(a+h)}$$

$$= -\frac{1}{g(a)} \cdot g'(a) \cdot \frac{1}{g(a)}$$

$$= -\frac{g'(a)}{[g(a)]^2}$$

$$(2) \left(\frac{f}{g}\right)'(a) = \left(f \cdot \frac{1}{g}\right)'(a) \quad *$$

Compositions If g diff. at a , f diff at $g(a)$, then $f \circ g$ is diff at a &

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a)$$

$$\begin{aligned} \text{pf WTS } \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{h} &\text{ exists.} \\ &= \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} \cdot \frac{g(a+h) - g(a)}{h} \end{aligned}$$

$$\text{WTS } \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} \text{ exists.}$$

Let $\varepsilon > 0$, want $\delta > 0$, s.t. whenever $0 < |h| < \delta$, we have

$$\left| \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} - f'(g(a)) \right| < \varepsilon$$

Since f is diff at $g(a)$.

$$\lim_{k \rightarrow 0} \frac{f(g(a+k)) - f(g(a))}{k} \text{ exists.}$$

If k is $g(a+h) - g(a)$

then $f(g(a+k)) = f(g(a+h))$

Since g is diff. at a , it is cts. at a .

$$\text{For } 0 < |h| < \textcircled{?1}^{\delta'}, |g(a+h) - g(a)| < \textcircled{?2}^{\delta}$$

$$\text{If } |k| < \textcircled{?3}^{\delta}, \text{ then } \left| \frac{f(g(a+k)) - f(g(a))}{k} - f'(g(a)) \right| < \textcircled{?4}^{\varepsilon}$$

For the given $\varepsilon > 0$, if $0 < |h| < \delta'$, then every limit that I've written down exists

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} \text{ exists.} \Rightarrow f'(g(a)) \cdot g'(a)$$